

SUB-QUERY:

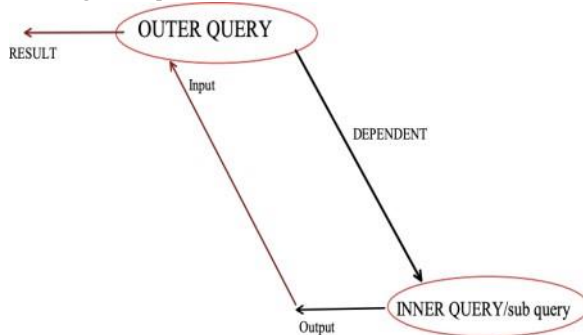
"A query written inside another query is known As sub query "

NESTED SUB QUERY:

“A sub query written inside a sub query is known as Nested Subquery”

○ WE CAN NEST ABOUT 255 SUB QUERIES

Working Principle :



Let us consider two queries Outer Query and Inner Query .

- Inner Query executes first and produces an Output .
- The Output of Inner Query is given / fed as an Input to Outer Query .
- The Outer Query generates the Result.
- Therefore we can state that 'the Outer Query is dependent on Inner Query' and this is the Execution Principle of Sub Query.

Why / When Do we use SUB QUERY :

Case 1:

Whenever we have **Unknowns present** in the Question, We use sub query to find The Unknown values

Case2

Mapping multiple tables by using common columns between them. When data to be fetched is in one table and condition is in the another table, then we will use case2 (without relationship is not able to map)

CASE-3

“MAPPING A TABLE ITSELF BY USING DIFFERENT COLUMNS OF THE TABLE BUT COLUMNS MUST BE ASSIGNED WITH SAME DATATYPES AND SIZE”

CASE-4

“To Find nth max or nth min value” (we need to n number of queries)

Case 1:

Whenever we have **Unknowns present** in the Question, We use sub query to find The Unknown values

Example :

EMP

<u>EID</u>	<u>ENAME</u>	<u>SAL</u>	<u>DEPTNO</u>
1	ALLEN	1000	20
2	BLAKE	2000	10
3	CLARK	3000	30
4	MILLER	1500	10
5	SMITH	2500	10

1. WAQTD names of the employees earning less than MILLER .

select ename from emp where sal < (select sal from emp where ename='miller');

select ename from emp where sal < 1500

NOTE:

- In the Inner Query / Sub Query we cannot select more than One column .
- The corresponding columns need not be same , but the datatypes of those has to be same .

Drawback- output is valid but not clear

SUB-QUERY

CASE-2:

Whenever the data to be selected and the condition to be executed are present in different tables we use Sub Query .

Example :

EMP

<u>EI</u> <u>D</u>	<u>ENAME</u>	<u>SAL</u>	<u>DEPTNO</u>
1	ALLEN	1000	20
2	BLAKE	2000	10
3	CLARK	3000	30
4	MILLER	1500	10
5	ADAMS	2500	20

DEPT

<u>DEPTNO</u>	<u>DNAME</u>	<u>LOC</u>
10	D1	L1
20	D2	L2
30	D3	L3

1. WAQTD **dname** of the employee whose name is **Miller** .

```

SELECT DNAME
FROM DEPT
WHERE DEPTNO= (SELECT DEPTNO
                FROM EMP
                WHERE ENAME='MILLER');
```

2. WAQTD Location of ADAMS

```

SELECT LOC
FROM DEPT
WHERE DEPTNO =(SELECT DEPTNO
                FROM EMP
                WHERE ENAME='ADAMS');
```

EXAMPLE 2:**CUSTOMER****PRODUCT**

<u>CID</u>	<u>CNAME</u>	<u>CNO</u>	<u>PID</u>	<u>PID</u>	<u>PNAME</u>	<u>PRICE</u>	<u>DISCOUNT</u>
1	SMITH	12345	101	101	Iphone 11	51000	0
2	JONES	12349	102	102	Iphone 12	74000	1000
3	ALLEN	12346	103	103	iPad	40000	100

3. WATQD the name of the product that smith purchased .

```

SELECT PNAME
FROM PRODUCT
WHERE PID =(SELECT PID
            FROM CUSTOMER
            WHERE CNAME='SMITH');
```

Drawback-we can map multiple tables but we cannot fetch multiple table Data

TYPES OF SUB - QUERY :

- 1) SINGLE ROW SUB QUERY
- 2) MULTI ROW SUB QUERY

EXAMPLE:

EMP

EID	ENAME	SAL	DEPTNO
1	ALLEN	1000	20
2	BLAKE	2000	10
3	CLARK	3000	30
4	MILLER	1500	10
5	SMITH	2500	10

DEPT

DEPTNO	DNAME	LOC
10	D1	L1
20	D2	L2
30	D3	L3

† SINGLE ROW SUB QUERY:

- If the sub query returns exactly 1 record / value we call it as Single Row Sub Query.
- *If it returns only 1 value then we can use the normal operators Or the Special Operators to compare the values.*

EXAMPLE

- WAQTD DNAME OF ALLEN
SELECT DNAME

```
FROM DEPT
WHERE DEPTNO =(SELECT DEPTNO
                FROM EMP
                WHERE ENAME='ALLEN');
```

‡ MULTI ROW SUB QUERY:

- If the sub query returns more than 1 record / value we call it as Multi Row Sub Query.
- If it returns more than 1 value then we **cannot use the normal operators** We have to **use only Special Operators** to compare the values.

EXAMPLE

- WAQTD DNAME OF ALLEN OR SMITH.

```
SELECT DNAME
FROM DEPT
```

```
WHERE DEPTNO = (SELECT
DEPTNO
FROM
EMP
WHERE ENAME IN('ALLEN' ,
'SMITH'));
```

query returns 2
records / values so we cannot use
'=' op.
we have to use **'IN'** OP

```
SELECT DNAME
FROM DEPT
WHERE DEPTNO IN (SELECT DEPTNO
FROM EMP
WHERE ENAME IN('ALLEN' ,
'SMITH'));
```

Note :

It is difficult to identify whether a query Belongs Single or Multi row So , it is always recommended to use Special Operators to Compare The values.

1. WAQTD Ename and salary of the employees earning more than Employees of dept 10.

EID	ENAME	SAL	DEPTNO
1	ALLEN	1000	20
2	BLAKE	2000	10
3	CLARK	3000	30
4	MILLER	1500	10
5	SMITH	2500	10

EMP

SELECT ENAME ,SAL
FROM EMP

WHERE SAL > (SELECT
SAL

2000
1500
2500

FROM EMP
WHERE

DEPTNO= 10);

HERE we cannot use > symbol to
compare multiple values

we cannot use IN or NOT IN OP as
well because it is used for = and
<> symbol

therefore we have to use SUB
QUERY Operator to comparing
relational operators such as (>,<,<=,>=)

Sub Query Operators :

1. ALL
2. ANY
3. EXIST
4. NOT EXIST

† ALL :

"It is special Op used along with a relational Op (> , < , > = , <=) to compare the values present at the RHS".

ALL Op returns true if all the values at the RHS have satisfied the condition .

EAXMPLE

1. WAQTD Ename and salary of the employees earning more than Employees of dept 10.

SELECT ENAME,SAL (CLARK,
3000) FROM EMP
WHERE SAL > ALL (SELECT
SAL

2000
1500
2500

FROM

EMP

WHERE

DEPTNO =10);

†	1000		
†	2000		
†	3000	† REJECTED	
†	1500	1000 > 2000	FALSE
†	2500	1000 > 1500	FALSE
		1000 > 2500	FALSE
		2000 > 2000	F
		2000 > 1500	T
		2000 > 2500	F

REJECTED

**SELECT ENAME,SAL (BLAKE ,CLARK ,SMITH
2000,3000,2500)**

FROM EMP

WHERE SAL > ANY

**(SELECT SAL
FROM EMP
WHERE DEPTNO=10);**

1000
2000
3000
1500
2500

2000
1500
2500

1000>2000	F
1000>1500	F
1000 > 2500	F
REJECTED	
2000>2000	F
2000>1500	T
2000>2500	F
SELECETD	

3000>2000	T
3000> 1500	T
3000>2500	T
SELECTED	
1500>2000	F
1500>1500	F
1500> 2500	F
REJECTED	

2500>2000	T
2500>1500	T
2500>2500	F
SELECTED	

SUB-QUERY

CASE-3

“MAPPING A TABLE ITSELF BY USING DIFFERENT COLUMNS OF THE TABLE BUT COLUMNS MUST BE ASSIGNED WITH SAME DATATYPES AND SIZE”

EMPLOYEE AND MANAGER RELATION:

EMP

<u>EMPNO</u>	<u>ENAME</u>	<u>MGR</u>
1	ALLEN	3

2	SMITH	1
3	JAMES	2
4	KING	3

NOTE:

To find Manager/Reporting mgr	Select MGR in Sub Q
To find Employees	Select EMPNO in Sub Q

CASE 1: TO FIND MANAGER

1. WAQTD name of Allen's manager

```
SELECT ENAME (JAMES)
FROM EMP
WHERE EMPNO= (SELECT MGR (3)
               FROM EMP
               WHERE ENAME='ALLEN');
```

2. WAQTD name of SMITH's manager .

```
SELECT ENAME
FROM EMP
WHERE EMPNO IN (SELECT MGR
                 FROM EMP
                 WHERE ENAME='SMITH');
```

3. WAQTD Names of smiths manager's manager. SELECT

```
ENAME
FROM EMP
WHERE EMPNO IN(SELECT MGR
                FROM EMP
                WHERE EMPNO IN(SELECT MGR
                                FROM EMP
                                WHERE
                                ENAME='SMITH'));
```

4. WAQTD dname of King's Manager.

```

SELECT DNAME
FROM DEPT
WHERE DEPTNO IN (SELECT DEPTNO
                  FROM EMP
                  WHERE EMPNO IN (SELECT MGR
                                  FROM EMP
                                  WHERE
ENAME='KING')));

```

5. WAQTD Location of Adams's manager's manager .

```

SELECT LOC
FROM DEPT
WHERE DETPNO IN (SELECT DEPTNO
                 FROM EMP
                 WHERE EMPNO IN(SELECT MGR
                                FROM EMP
                                WHERE EMPNO IN(SELECT MGR
                                                FROM EMP
                                                WHERE ENAME='ADAMS'))));

```

CASE 2: TO FIND EMPLOYEES

1. WAQTD Names of the employees reporting to KING.

```

SELECT ENAME
FROM EMP
WHERE MGR IN (SELECT EMPNO
              FROM EMP
              WHERE ENAME='KING');

```

2. WAQTD Name and salary given to the employees reporting To James .

```

SELECT ENAME,SAL
FROM EMP
WHERE MGR IN (SELECT EMPNO
              FROM EMP
              WHERE ENAME='JAMES');

```

3. WAQTD dname of the employee reporting to President .

4. WAQTD Department details of the employees who are reporting to MILLER .

```
SELECT *
FROM DEPT
WHERE DEPTNO IN (SELECT DEPTNO
FROM EMP
WHERE MGR IN (SELECT EMPNO
FROM EMP
WHERE ENAME='MILLER'));
```